

GYNOGEN HP 75 IU/150 IU

**Powder and Solvent for Solution for Injection
(Menotrophin for Injection BP, 75 IU / Vial & 150 IU / Vial)**

For Intramuscular / Subcutaneous Injection only
Highly Purified & Lyophilized Powder for Injection

DESCRIPTION :

Each Vial of sterile freeze-dried product contains :
Menotrophin.....BP
Equivalent to the activity of
Follicle Stimulating Hormone (FSH).....75 IU
Luteinizing Hormone (LH).....75 IU

Each Vial of sterile freeze-dried product contains :
Menotrophin.....BP
Equivalent to the activity of
Follicle Stimulating Hormone (FSH).....150 IU
Luteinizing Hormone (LH).....150 IU

Product: An almost white or slightly yellowish freeze-dried product sealed in a 2mL USP Type-I clear glass vial

Reconstitution solution: A clear, colourless solution is filled and duly sealed in a 2 mL USP Type-I clear glass ampoule.

After reconstitution: Clear colourless solution.
The product **GYNOGEN HP 75/150 IU** is compatible with the reconstitution solutions i.e. 0.9%w/v sodium chloride injection.

Each vial with powder contains human menopausal gonadotrophin, HMG corresponding to 75/150 IU human follicle stimulating hormone activity (FSH) and 75/150 IU human luteinizing hormone activity (LH).

PHARMACOLOGY: PHARMACODYNAMICS:

Menotrophin is a gonadotrophin extracted from the urine of postmenopausal women and having both luteinising hormone and follicle stimulating hormone activity. It is given by intramuscular or subcutaneous injection in the treatment of male and female infertility.

Menotrophin (HMG) directly affects the ovaries and the testes.HMG has a gametropic and steroidogenic effect.

In the ovaries, FSH-component in HMG induces an increase in the number of growing follicles and stimulates their development. FSH increases the production of oestradiol in the granulosa cells by aromatising androgens that originate in the Theca cells under the influence of the LH-component.

In the testes, FSH induces the transformation of premature to mature Sertoli cells. It mainly causes the maturation of the seminal canals and the development of the spermatozoa. However, a high concentration of androgens within the testes is necessary and can be attained by a prior treatment using hCG.

PHARMACOKINETICS:

HMG is not effective when taken orally and is injected either intramuscularly or subcutaneously. The biological effectiveness of HMG is mainly due to its FSH content. The pharmacokinetics of HMG following intramuscular or subcutaneous administration show great individual variation. The maximum serum level of FSH is reached approximately 18 hours after intramuscular injection and 12 hours after subcutaneous injection. After that, the serum level decreases by a half-life of approximately 55 hours following intramuscular administration and 50 hours following subcutaneous administration.Excretion of HMG, following administration, is predominantly renal.

INDICATIONS

GYNOGEN HP is indicated for the treatment of infertility in the following clinical situations:

Anovulation, including polycystic ovarian disease (PCOD), in women who have been unresponsive to treatment with clomiphene citrate.

Controlled ovarian hyperstimulation to induce the development of multiple follicles for assisted reproductive technologies (ART) (e.g. in vitro fertilization/embryo transfer (IVF/ET), gamete intra-fallopian transfer (GiFT) and intracytoplasmic sperm injection (ICSI).

DOSAGE & ADMINISTRATION

Treatment with **GYNOGEN HP** should be initiated under the supervision of a physician experienced in the treatment of fertility problems.

Method of administration: Dosage regimens described below are identical for subcutaneous (S.C.) and intramuscular (I.M.) administration. **GYNOGEN HP** is intended for subcutaneous (S.C.) or intramuscular (I.M.) injection after reconstitution with the solvent provided.

The powder should be reconstituted immediately prior to use. In order to avoid the injection of large volumes up to 3 vials of the powder may be dissolved in 1 ml of the solvent provided.

Shaking should be avoided. The solution should not be used if it contains particles or if it is not clear.

Dosage: There are great inter-individual variations in the response of the ovaries to exogenous gonadotrophins. This makes it impossible to set a uniform dosage scheme. The dosage should, therefore, be adjusted individually depending on the ovarian response. **GYNOGEN HP** can be given alone or in combination with a gonadotrophin-releasing hormone (GnRH) agonist or antagonist. Recommendations about dosage and duration of treatment may change depending on the actual treatment protocol.

Women with anovulation (including PCOD): The object of **GYNOGEN HP** therapy is to develop a single Graafian follicle from which the oocyte will be liberated after the administration of human chorionic gonadotrophin (hCG).

GYNOGEN HP therapy should start within the initial 7 days of the menstrual cycle. The recommended initial dose of GYNOGEN HP is 75-150 IU daily, which should be maintained for at least 7 days. Based on clinical monitoring (including ovarian ultrasound alone or in combination with measurement of oestradiol levels) subsequent dosing should be adjusted according to individual patient response. Adjustments in dose should not be made more frequently than every 7 days. The recommended dose increment is 37.5 IU per adjustment, and should not exceed 75 IU. The maximum daily dose should not be higher than 225 IU. If a patient fails to respond adequately after 4 weeks of treatment, that cycle should be abandoned and the patient should recommence treatment at a higher starting dose than in the abandoned cycle.

When an optimal response is obtained, a single injection of 5,000 IU to 10,000 IU hCG should be given 1 day after the last GYNOGEN HP injection. The patient is recommended to have coitus on the day of and the day following hCG administration. Alternatively intrauterine inseminat (IUI) may be performed. If an excessive response to GYNOGEN HP is obtained treatment should be stopped and hCG withheld (see Special warning and precautions for use) and the patient should use a barrier method of contraception or refrain from having coitus until the next menstrual bleeding has started.

Women undergoing controlled ovarian hyperstimulation for multiple follicular development for assisted reproductive technologies (ART):

In a protocol using down-regulation with a GnRH agonist, GYNOGEN HP therapy should start approximately 2 weeks after the start of agonist treatment. In a protocol using down-regulation with a GnRH antagonist, GYNOGEN HP therapy should start on day 2 or 3 of the menstrual cycle. The recommended initial dose of GYNOGEN HP is 150-225 IU daily for at least the first 5 days of treatment. Based on clinical monitoring (including ovarian ultrasound alone or in combination with measurement of oestradiol levels) subsequent dosing should be adjusted according to individual patient response, and should not exceed more than 150 IU per adjustment. The maximum daily dose given should not be higher than 450 IU daily and in most cases dosing beyond 20 days is not recommended.

When a suitable number of follicles have reached an appropriate size a single injection of up to 10,000 IU hCG should be administered to induce final follicular maturation in preparation for oocyte retrieval. Patients should be followed closely for at least 2 weeks after hCG administration. If an excessive response to GYNOGEN HP is obtained treatment should be stopped and hCG withheld (see section Special warnings and precautions for use) and the patient should use a barrier method of contraception or refrain from having coitus until the next menstrual bleeding has started.

Renal/hepatic impairment: Patients with renal and hepatic impairment have not been included in clinical trials (see section Pharmacokinetics).

Paediatric population: There is no relevant use of **GYNOGEN HP** in the paediatric population.

CONTRAINDICATIONS

Men and Women

GYNOGEN HP is contraindicated in men and women with:

- Tumours of the pituitary or hypothalamic glands
- Hypersensitivity to the active substance or any of the excipients used in the formulation.

Men

- Tumours in the testes.
- Prostate carcinoma.

Women

- Ovarian, uterine or mammary carcinoma.
- Pregnancy and lactation.
- Gynaecological haemorrhage of unknown aetiology.
- Ovarian cysts or enlarged ovaries not due to polycystic ovarian disease.

In the following situations treatment outcome is unlikely to be favourable, and therefore **GYNOGEN HP** should not be administered:

- Primary ovarian failure.
- Malformation of sexual organs incompatible with pregnancy.
- Fibroid tumours of the uterus incompatible with pregnancy.
- Structural abnormalities in which a satisfactory outcome cannot be expected, for example, tubal occlusion (unless superovulation is to be induced for IVF), ovarian dysgenesis, absent uterus or premature menopause.

WARNINGS & PRECAUTIONS

GYNOGEN HP is a potent gonadotropic substance capable of causing mild to severe adverse reactions, and should only be used by physicians who are thoroughly familiar with infertility problems and their management.

In the treatment of female infertility, ovarian activity should be checked (by ultrasound and plasma 17 beta-oestradiol measurement) prior to **GYNOGEN HP** administration. During treatment, these tests and urinary oestrogen measurement should be carried out at regular intervals, until stimulation occurs. Close supervision is imperative during treatment. See "posology and administration" for optimum response levels of urinary oestrogen and plasma 17 beta-oestradiol. Values below these ranges may indicate inadequate follicular development.

There is considerable inter-patient variability in response to menotrophin administration, with a poor response to menotrophin in some patients. The lowest effective dose in relation to the treatment objective should be used.The first injection of **GYNOGEN HP** should be performed under direct medical supervision.

Before starting treatment, the couple's infertility should be assessed as

appropriate and putative contraindications for pregnancy evaluated. In particular, patients should be evaluated for hypothyroidism, adrenocortical deficiency, hyperprolactinemia and pituitary or hypothalamic tumours, and appropriate specific treatment given.

Patients undergoing stimulation of follicular growth, whether in the frame of a treatment for anovulatory infertility or ART procedures may experience ovarian enlargement or develop hyperstimulation. Adherence to recommended **GYNOGEN HP** dosage and regimen of administration, and careful monitoring of therapy will minimise the incidence of such events. Acute interpretation of the indices of follicle development and maturation requires a physician who is experienced in the interpretation of the relevant tests.

Ovarian Hyperstimulation Syndrome (OHSS)

OHSS is a medical event distinct from uncomplicated ovarian enlargement. OHSS is a syndrome that can manifest itself with increasing degrees of severity. It comprises marked ovarian enlargement, high serum sex steroids, and an increase in vascular permeability which can result in an accumulation of fluid in the peritoneal, pleural and rarely, in the pericardial cavities.

The severe form OHSS may be life-threatening and is characterised by large ovarian cysts (prone to rupture), acute abdominal pain, ascites, very often hydrothorax and occasionally thromboembolic phenomena. Other symptoms that may be observed include: abdominal distension, severe ovarian enlargement, weight gain, dyspnoea, oliguria and gastrointestinal symptoms including nausea, vomiting and diarrhoea. Clinical evaluation may reveal hypovolaemia, haemoconcentration, electrolyte imbalances, haemoperitoneum, pleural effusions and acute pulmonary distress.

If urinary oestrogen levels exceed 540 nmol (150 micrograms)/24 hours, or if plasma 17 beta-oestradiol levels exceed 3000 pmol/L (800 picograms/ml), or if there is any steep rise in values, there is an increased risk of hyperstimulation and **GYNOGEN HP** treatment should be immediately discontinued and human chorionic gonadotrophin withheld. Ultrasound will reveal any excessive follicular development and unintentional hyperstimulation. In the event of hyperstimulation, the patient should refrain from sexual intercourse or to use barrier contraception methods for at least 4 days. OHSS may progress rapidly (within 24 hours to several days) to become a serious medical event, therefore patients should be followed for at least two weeks after the hCG administration.

If during ultrasound, several mature follicles are visualised, human chorionic gonadotrophin should not be given as there is a risk of multiple ovulation and the occurrence of hyperstimulation syndrome.

Adherence to recommended **GYNOGEN HP** dosage, regimen of administration and careful monitoring of therapy will minimise the incidence of ovarian hyperstimulation and multiple pregnancy. Patients undergoing superovulation may be at an increased risk of developing hyperstimulation in view of the excessive oestrogen response and multiple follicular development. In ART, aspiration of all follicles prior to ovulation may reduce the occurrence of hyperstimulation.

OHSS may be more severe and more protracted if pregnancy occurs. Most often, OHSS occurs after hormonal treatment has been discontinued and reaches its maximum at about seven to ten days following treatment. Usually, OHSS resolves spontaneously with the onset of menses. If severe OHSS occurs, gonadotrophin treatment should be stopped if still ongoing, the patient hospitalised and specific therapy for OHSS started. This syndrome occurs with higher incidence in patients with polycystic ovarian disease.

Multiple pregnancy

Multiple pregnancy, especially high order, carries an increased risk of adverse maternal and perinatal outcomes.

In patients undergoing ART procedures the risk of multiple pregnancy is related mainly to the number of embryos replaced, their quality and the age of the patient. The patient should be advised of the potential risk of multiple births before starting treatment.

Pregnancy wastage

The incidence of pregnancy wastage by miscarriage or abortion is higher in patients undergoing stimulation of follicular growth for ART procedures than in the normal population.

Ectopic pregnancy

Women with a history of tubal disease are at risk of ectopic pregnancy, whether the pregnancy is obtained by spontaneous conception or with fertility treatment. The prevalence of ectopic pregnancy after IVF has been reported to be 2 to 5%, as compared to 1 to 1.5% in the general population.

Reproductive system neoplasms

There have been reports of ovarian and other reproductive system neoplasms, both benign and malignant, in women who have undergone multiple drug regimens for infertility treatment. It is not yet established if treatment with gonadotrophins increases the baseline risk of these tumours in infertile women.

Congenital malformation

The prevalence of congenital malformations after ART may be slightly higher than after spontaneous conceptions. This is thought to be due to differences in parental characteristics (e.g. maternal age, sperm characteristics) and multiple pregnancies.

Thromboembolic events

In women with generally recognised risk factors for thromboembolic events, such as personal or family history, treatment with gonadotrophins may further increase the risk. In these women, the benefits of gonadotrophin administration need to be weighed against the risks. It should be noted however, that pregnancy itself also carries an increased risk of thromboembolic events.